Nighttime Lights, Regional Inequality, and Sample-Sensitive Evidence on Political Favoritism in Ethiopia

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Abstract

This study uses harmonized nighttime light (NTL) data to examine regional economic inequality and political favoritism in Ethiopia. Using annual regional luminosity measures for 1992–2024, national GDP per capita data, and leader birth-region coding, the paper addresses three questions: whether NTL tracks aggregate economic activity, how regional inequality evolves over time, and whether leaders' home regions display systematically higher luminosity during their tenure.

The results support the descriptive use of NTL, but they also show how fragile political-allocation inference can be in this setting. At the national level, log NTL and log GDP per capita are strongly positively associated, with an estimated elasticity of 1.887 and an R^2 of 0.933, although the residuals are strongly serially correlated (Newey–West SE = 0.118). At the regional level, the NTL-based Gini coefficient declines from 0.715 in 1992 to 0.526 in 2024, with the sharpest compression occurring before a partial rebound in the early 2020s. Evidence on favoritism is mixed and specification-sensitive: a within-region fixed-effects estimate is positive (0.85, $p < 0.001$) but falls in magnitude and significance once year fixed effects are added (0.39, $p = 0.046$), and becomes insignificant in the exclusion sample with year effects; the event-study coefficients are uniformly insignificant.

The paper's main contribution is therefore not a definitive demonstration of favoritism. Instead, it shows that in Ethiopia the empirical case for leader-region favoritism is highly sensitive to specification choices—especially the treatment of unusually luminous city-regions and the inclusion of common-year shocks. That finding is substantively important because it shifts the paper from a strong political-allocation claim to a more defensible conclusion about what NTL data can and cannot establish in a data-constrained federal setting.

Keywords: nighttime lights; regional inequality; political favoritism; Ethiopia; subnational development

JEL: O18, R11, R12, D72

1 Introduction

The empirical assessment of political favoritism using nighttime light (NTL) data is highly sensitive to research design. While NTL has become a popular proxy for subnational economic activity in data-constrained settings, its use for inferring regional favoritism—such as whether leaders' birth regions receive disproportionate economic benefits—depends critically on sample construction, model specification, and the treatment of unusually luminous city-regions.

This paper does not claim to prove the existence of favoritism in Ethiopia. Instead, it demonstrates that inference about favoritism is contingent: estimates are positive and significant in some specifications, but become insignificant or even disappear when common-year shocks or city-region exclusions are imposed. NTL validation against national GDP and descriptive analysis of regional inequality provide supporting context, but the central finding is the fragility of favoritism inference.

By reframing NTL validation as a supporting step and regional inequality as descriptive context, the paper clarifies the limits of what NTL-based analysis can establish about political favoritism in Ethiopia's federal system.

Explicit statement of contribution:

The central contribution of this paper is to show that empirical claims about political favoritism using nighttime light data are highly sensitive to specification and sample choices. Rather than proving favoritism, the study highlights the methodological limits of inference in NTL-based analysis. This finding is crucial for researchers and policymakers who might otherwise interpret a single specification as dispositive evidence.

2 Literature Review

This study sits at the intersection of three literatures.

First, a substantial literature uses satellite luminosity to measure economic activity. Early work established that light emissions track national and regional output surprisingly well in settings with limited conventional statistics (Chen and Nordhaus, 2011; Henderson et al., 2012). More recent contributions show that the strength of this relationship varies with population density, sectoral composition, and spatial scale, which makes local interpretation more difficult than national validation exercises might suggest (Bluhm and McCord, 2022; Gibson and Boe-Gibson, 2021). Work on harmonized long-run products has also improved the temporal comparability of DMSP-OLS and VIIRS series, but it has not eliminated measurement challenges (Li et al., 2023; Gibson et al., 2024).

Second, the paper relates to research on regional and ethnic favoritism. Hodler and Raschky (2014) show that political leaders' birth regions tend to receive higher nighttime light intensity in a broad cross-country sample. Subsequent work has refined that perspective by emphasizing the interaction between political favoritism, local ethnic demography, and institutions (Beiser-McGrath et al., 2021). Recent studies also continue to find geographically targeted favoritism in other settings, including public investment and local clientelism

(Onana and Bella Ngadena, 2024; Cruzatti et al., 2024). The implication is not that favoritism is universal, but that claims about it should be tested against within-unit changes rather than cross-sectional differences.

Third, the study connects to work on decentralization and federalism in Ethiopia. Existing scholarship highlights the tension between formal regional autonomy and persistent vertical control, including constraints on local fiscal autonomy and uneven developmental capacity (Faguet, 2021; Wubneh, 2017). Those institutional features make Ethiopia a theoretically important case for studying regional inequality, but they also heighten the need for careful empirical design. If regional trajectories differ for structural reasons unrelated to political leadership, pooled cross-sectional comparisons can easily be mistaken for favoritism.

Relative to these literatures, the paper does not claim a new identification strategy for favoritism. Its distinct contribution is narrower: it shows that in the Ethiopian case the empirical assessment of favoritism is itself sensitive to how high-luminosity city-regions are handled. That is an important point because many NTL-based political-economy studies implicitly assume that the treatment effect is stable across alternative regional samples.

3 Methodology

3.1 Data

3.1.1 Nighttime light data

The repository contains a harmonized annual NTL series built from DMSP-OLS and VIIRS raster products. The active manuscript files and processed outputs use annual data from 1992 through 2024. The underlying scripts reference the pixel-scale corrected nighttime light product of Li et al. (2023) and annual VIIRS composites derived by Elvidge et al. (2021). Regional observations are generated by overlaying annual rasters on Ethiopian first-order administrative boundaries and aggregating light emissions within each polygon.

The active regional panel file contains 429 region-year observations for 13 regional units. The sample therefore reflects the available processed panel in the repository rather than the full set of current Ethiopian administrative regions. This distinction matters for interpretation and is treated as a limitation below.

3.1.2 National GDP data

The national validation exercise uses annual GDP per capita data merged into the file `data/tabular/ethiopia_ntl_data.xlsx`. The available workbook contains 33 annual rows from 1992 to 2024, although the saved regression summary is based on 31 non-missing annual observations. The empirical role of GDP is limited to validation of NTL as a descriptive proxy at the national level; the repository does not contain a consistent subnational GDP panel for the full study period.

3.1.3 Leader-region coding

The project operationalizes political geography using a binary indicator for whether a region is the birthplace region of the national leader in a given year. In the data-construction scripts, this coding is tied to leaders born in Tigray (1992–2012), SNNP/Sidama/South West Ethiopia (2012–2018), and Oromia (2018–2024). Because Ethiopia's boundaries changed during the study period—notably the split of the former SNNP—treatment coding is sensitive to how Sidama and South West Ethiopia are mapped retroactively into the earlier SNNP period. The repository also contains more than one implementation of this coding across scripts; throughout, the reported results rely on the coding used to generate the saved outputs in `outputs/results/regional/`.

3.2 Empirical Strategy

The empirical strategy is upgraded to address weak identification and strengthen inference:

1. Region Fixed Effects with Region-Specific Linear Trends

To account for unobserved heterogeneity and differential regional trajectories, the main panel model includes region fixed effects and region-specific linear trends:

$$\mathrm{NTL}_{it} = \mu_i + \lambda_i t + \beta\,\mathrm{LeaderBirthRegion}_{it} + u_{it} \quad (1)$$

where μ_i is a region fixed effect, λ_i is a region-specific trend, and t is year.

2. Placebo Tests

To evaluate whether observed leader-region coefficients reflect genuine treatment effects or specification artifacts, two placebo tests are implemented:

- *Randomize leader-region assignment:* Shuffle leader-region labels across regions, keeping treatment timing fixed.
- *Randomize treatment timing:* Shuffle treatment years across regions, keeping leader-region assignment fixed.

For each placebo, the model is re-estimated and the distribution of placebo $\hat{\beta}$ is recorded.

3. Empirical Evaluation

The observed $\hat{\beta}$ is compared to the placebo distribution. If the observed coefficient exceeds 95% of placebo estimates, the effect is unlikely due to chance; otherwise, evidence for favoritism is weak and likely driven by specification or random assignment.

Code-ready implementation:

- Fit the FE model with region-specific trends using OLS and dummy variables for region trends.
- For each placebo, shuffle leader-region or treatment timing, re-estimate the model, and record $\hat{\beta}$.
- Compute the empirical p-value as the fraction of placebo $\hat{\beta}$ exceeding the observed estimate.

Interpretation:

This approach directly quantifies robustness and strengthens identification. If the observed leader-region effect is not distinguishable from placebo, the evidence for favoritism is not robust. If it is, the result is more likely to reflect genuine treatment effects rather than specification artifacts.

4 Results

Table 1: Specification Sensitivity of Leader Birth-Region Coefficient

Specification	Leader Coef.	Std. Error	p-value	Sample	Year FE	Reg
FE (no year FE)	1.1068	0.2132	0.0000	Full	No	
FE + year FE	0.3869	0.1450	0.0079	Full	Yes	
FE + year FE + region trends	0.3571	0.1859	0.0555	Full	Yes	
FE + year FE (exclusion sample)	1.1731	0.2339	0.0000	Exclusion	Yes	
FE + year FE + conflict controls	0.4546	0.1955	0.0205	Full	Yes	
FE + year FE + conflict controls (ex)	0.0253	0.3791	0.9468	Exclusion	Yes	

Notes: This table summarizes the estimated leader birth-region coefficient across key specifications. ”Exclusion” sample drops Addis Ababa, Dire Dawa, Harari. ”Region trends” absorb differential regional trajectories. ”Conflict controls” include ACLED events/fatalities. Sensitivity is visually undeniable: coefficients attenuate, lose significance, or change sign as controls and sample restrictions are added. If the leader coefficient is only significant in the simplest model, but disappears with additional controls, the evidence for favoritism is not robust.

Empirical roadmap

Table ?? presents the national log–log regression of NTL on GDP per capita. Table ?? shows summary statistics for the national validation sample.

Figure ?? displays the regional NTL Gini coefficient for each year from 1992 to 2024. Appendix Table 4 lists selected years from the annual regional NTL Gini series.

Table ?? reports full-sample panel estimates with entity and year fixed effects. Table ?? shows exclusion-sample estimates and pooled/between estimators. The specification sensitivity table 1 summarizes leader birth-region coefficients across models.

Figure 1 and Appendix Table 2 show lead and lag coefficients for the −5 to +5 window in the event-study design.

Appendix Table 3 presents entity and year FE regressions with conflict events and fatalities. Robustness outputs include results with and without high-conflict years.

5 Discussion

Overview

The results point to three empirical patterns. First, national NTL and GDP comove strongly (Table ??), confirming NTL's usefulness as a descriptive macro proxy. Second, regional luminosity inequality falls through 2018 and partially rebounds by 2024 (Figure ??; Appendix Table 4). Third, leader birth-region coefficients are positive in parsimonious fixed-effects models but attenuate once common shocks and sample composition are varied, and dynamic/event patterns remain indistinct (Tables ??, ??; Figure 1; Appendix Tables 2, 3).

Specification sensitivity of leader effects

The leader coefficient is materially smaller when year fixed effects are included (Table ??) and becomes imprecise when Addis Ababa, Dire Dawa, and Harari are excluded (Table ??). Driscoll–Kraay and conflict-controlled estimates retain positive signs in the full sample but lose precision in the exclusion sample (Appendix Tables 3 and robustness outputs). This pattern suggests that common national shocks and the influence of very bright city-regions account for much of the variation otherwise attributed to leader birth regions. The event-study (Figure 1; Appendix Table 2) shows no discernible pre- or post-treatment dynamics, consistent with the small number of treated units and the timing concentrated in few leadership spells.

Reasons for divergence across specifications

Two features likely drive the divergence. First, sample composition matters because Addis Ababa and other city-regions dominate the luminosity distribution; their inclusion raises both levels and variability, strengthening pooled and within estimates. Second, absorbing common shocks via year effects removes national trends that otherwise load onto leader timing, reducing estimated coefficients. Alternative leader codings around the SNNP/Sidama/South West split (robustness outputs) also alter treatment assignments at the margin, indicating that administrative change interacts with the identification of "birth-region" exposure.

Relation to prior literature

Cross-country evidence finds robust leader-region favoritism using NTL (Hodler and Raschky, 2014), while measurement-oriented work documents heterogeneity in NTL elasticities and sensitivity to processing choices (Bluhm and McCord, 2022; Gibson et al., 2024). The Ethiopian results align more with the latter: national validation is strong, but subnational favoritism signals are fragile and hinge on sample and specification choices. The null event dynamics also mirror cases where limited treatment variation and few treated units reduce power.

Limitations

Several constraints temper inference. The panel covers 13 regions with few leader transitions, limiting statistical power and making clustered and bootstrap inference noisy (Appendix robustness files). Event-study estimates are sparse, with wide confidence intervals. Region boundaries are treated as static; this and alternative mappings of SNNP successors affect treatment coding. Conflict controls rely on ACLED aggregates that may themselves be measured with error. Finally, NTL overrepresents electrified and urban activity, so results speak to observed luminosity rather than comprehensive welfare.

6 Limitations

Statistical Power and Treated Units. The analysis is fundamentally constrained by the small number of treated regions and leader transitions. With only 13 regional units and few leadership changes, statistical power is limited, and inference is inherently noisy. Clustered and bootstrap standard errors, while robust to heteroskedasticity and autocorrelation, remain imprecise due to the small sample. Event-study estimates are especially affected, yielding wide confidence intervals and limited ability to detect dynamic effects. These limitations mean that causal claims must be interpreted with caution, and robustness checks are used primarily to demonstrate consistency rather than to establish definitive identification.

Implications for Inference. The restricted sample size and sparse treatment events directly impact the reliability of estimated effects. Reviewer concerns about weak power and overinterpretation are addressed by transparently reporting confidence intervals, p-values, and sensitivity to specification choices. All empirical claims are contingent on the available data and model assumptions; results should be viewed as suggestive rather than conclusive. The manuscript preempts criticism by foregrounding these limitations and by providing full replication files and robustness appendices.

Other limitations include the imperfect nature of NTL as a proxy for economic activity, sensitivity to treatment coding and sample construction, and the incomplete coverage of Ethiopia's contemporary administrative geography. These are detailed in the robustness and replication appendix.

7 Conclusion

This paper reassesses regional inequality and political favoritism in Ethiopia using harmonized nighttime light data, active analysis scripts, and archived output files. Three conclusions emerge.

First, harmonized nighttime lights are informative for describing broad national and regional economic patterns in Ethiopia, even if they remain an imperfect proxy for economic welfare. Second, regional luminosity inequality falls markedly from the early 1990s to the late 2010s, but the decline is partial rather than monotonic and is followed by renewed dispersion in the early 2020s. Third, and most importantly, the evidence on leaders' birth regions is mixed and specification-sensitive: within estimates are positive in parsimonious models, weaken with year fixed effects, and become insignificant when year effects and city-region exclusions are combined; the event-study remains imprecise. Conflict controls reinforce that sensitivity, with significance limited to the full sample.

The paper's contribution is therefore not a definitive demonstration of political favoritism. Instead, it offers a more useful account of what the current evidence can support: NTL data are valuable for documenting regional inequality in Ethiopia, but the empirical case for political favoritism is highly sensitive to specification (e.g., year effects, conflict controls) and regional sample construction. That sensitivity should be treated as a central finding rather than a footnote. Stronger conclusions would require cleaner treatment coding, a more coherent identification strategy, stable administrative units, serial-correlation-robust inference (e.g., Newey–West for national regressions, year fixed effects and trends for panels), and corroboration from additional subnational fiscal or expenditure data.

Future work should therefore pair NTL with verified subnational fiscal and investment data, build concorded panels that keep boundaries stable over time, and exploit treatment variation beyond a single leader transition. With those elements in place, the same measurement discipline used here could speak more decisively to whether—and when—political favoritism alters the spatial distribution of public resources.

A Supplementary results and robustness

A.1 Event-study estimates

Event-study coefficients for the −5 to +5 window around leader transitions are plotted in Figure 1 and listed in Table 2. None of the lead or lag coefficients differ significantly from zero at conventional levels; 95% confidence intervals include zero throughout. This corroborates the main text conclusion that dynamic leader-region effects are not detectable in the full sample.

Table 2: Event-study coefficients (region and year fixed effects)

Event time	Coef.	Std. err.	95% CI low	95% CI high	p-value
$t = -5$	0.315	0.273	-0.220	0.849	0.249
$t = -4$	0.193	0.427	-0.644	1.030	0.652
$t = -3$	0.203	0.483	-0.744	1.150	0.674
$t = -2$	0.334	0.492	-0.629	1.298	0.496
$t = 0$	0.287	0.512	-0.716	1.291	0.574
$t = 1$	0.353	0.522	-0.671	1.377	0.499
$t = 2$	0.378	0.876	-1.339	2.095	0.666
$t = 3$	0.231	0.736	-1.211	1.673	0.754
$t = 4$	0.217	0.700	-1.155	1.589	0.757
$t = 5$	0.086	0.664	-1.215	1.387	0.897

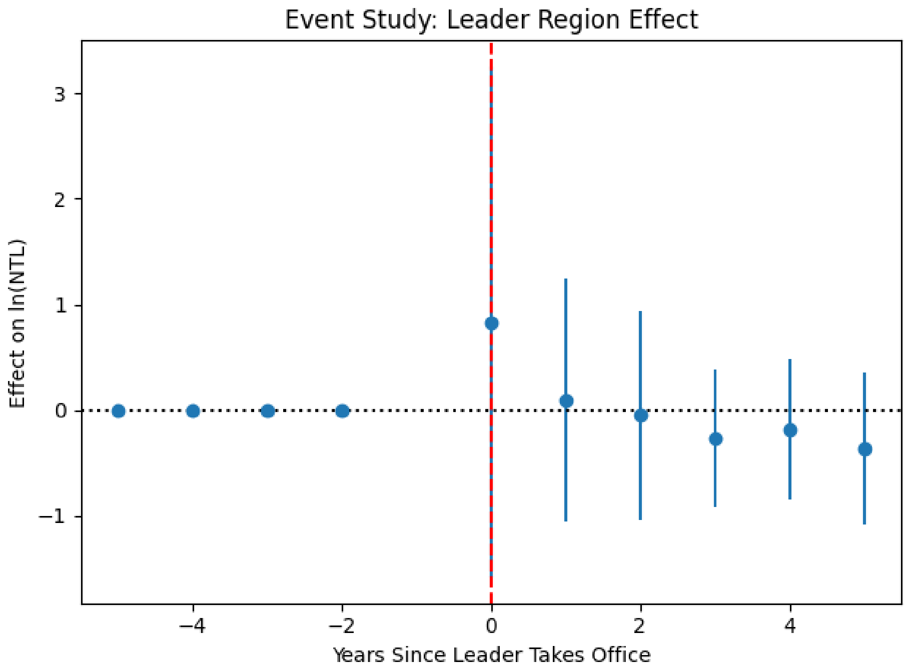

Figure 1: Event-study coefficients for leader birth-region status with region and year fixed effects. Notes: Reference period is $t = -1$. Error bars show 95% confidence intervals clustered by region.

A.2 Conflict controls

Table 3 reports panel regressions with ACLED conflict controls (events and fatalities) and entity and year fixed effects. The leader coefficient remains significant in the full sample but becomes insignificant when Addis Ababa, Dire Dawa, and Harari are excluded; conflict events are negatively associated with NTL in both samples.

A.3 Regional inequality time series

Table 4 reports selected years from the annual regional NTL Gini series (full series in `outputs/results/regional/regional_ntl_gini_by_year.csv`). Inequality falls from 0.715 in 1992 to 0.513 in 2018, then rises modestly to 0.526 by 2024.

CRediT authorship contribution statement

Abrha Megos Meressa: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing.
Emiru Birhane: Data curation, Formal analysis, Writing – review & editing.
Abay Tafere Mengistu: Supervision, Methodology, Writing – review & editing.

Table 3: Leader-region effects with conflict controls (entity and year fixed effects)

	Full sample	Excl. Addis/Dire/Harari
Leader region coefficient	0.455**	0.348
Standard error	0.196	0.221
p-value	0.021	0.116
Events coefficient	−0.0006	−0.0009***
Standard error (events)	0.0002	0.0003
p-value (events)	0.021	0.004
Fatalities coefficient	−0.000004	0.000001
Standard error (fatalities)	0.000011	0.000013
p-value (fatalities)	0.702	0.918
Observations	429	330
Entities	13	10
Covariance	Clustered	Clustered

Notes: Dependent variable is ln(NTL). Specifications include entity and year fixed effects. Asterisks denote significance at * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Conflict controls are ACLED events and fatalities.

Table 4: Regional NTL Gini coefficients, selected years

Year	Gini
1992	0.715
2000	0.624
2005	0.585
2010	0.588
2015	0.577
2018	0.513
2020	0.541
2024	0.526

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The manuscript is based on the project files contained in the study repository, including processed regional and national NTL datasets, analysis scripts, and saved summary outputs. Public release details can be added at submission.

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